Modules Software

Modules

Modules are a handy way to set up your environment for particular work, especially in a shared environment. They provide an easy way to load a particular version of a software, language, or compiler.

To see what modules are available, type the command:

```
module avail
```

Note

Engaging's newest hardware is running with the Rocky 8 operating system, whereas older hardware is using CentOS 7. You will see different module lists depending on whether you are on Rocky 8 or CentOS 7 node.

By default you will only see the modules for core and community software. All CentOS 7 software is considered deprecated.

To show deprecated modules on Rocky 8 load the `deprecated-modules` module:

```
module load deprecated-modules
```

To load a module, use the command:

```
module load moduleName
```

Where `moduleName` can be any of the modules listed by the `module avail` command.

> **Note**
>
> We do not recommend including `module load` commands in your `.bashrc`, `.bash_profile`, or any other startup scripts and instead include them in your job scripts. This provides a more predictable and consistent environment for your jobs. It is also very easy to forget that you have modules loaded in your `.bashrc`, and these can have impact on future workloads.

If you want to list the modules you currently have loaded, you can use the `module list` command:

```
module list
```

If you want to change to a different version of the module you have loaded, you can switch the module you have loaded. This is important to do when loading a different version of a module you already have loaded, the module command will not allow you to load two different versions of the same software. To switch modules run:

```
module switch oldModuleName newModuleName
```

where `oldModuleName` is the name of the module you currently have loaded, and `newModuleName` is the new module that you would like to load.

If you would like to unload the module, or remove the changes the module has made to your environment, use the following command:

```
module unload moduleName
```

If you would like to unload all modules in your environment, you can use the command:

```
module purge
```

This command is helpful when you want to ensure a clean environment. You can include it at the start of your job scripts to make sure your jobs all have a consistent environment. Loaded modules may carry over into your jobs, and sometimes can interfere with the work you are doing in unexpected ways.

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